

Rohan Kumar

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EDUCATION

University of Illinois at Urbana-Champaign

Expected Graduation Date: December 2026

Bachelor of Science in Computer Science and Statistics (GPA: 3.86 / 4.00)

Urbana-Champaign Area, IL

Coursework: Object Oriented Programming, Data Structures, Algorithms, Database Systems, Computer Systems, Distributed Systems, High Frequency Trading Technology, Algorithmic Market Microstructure, Statistical Learning, Machine Learning

SKILLS

Programming Languages: Python, C#, C/C++, Java, TypeScript, JavaScript, SQL, R, HTML/CSS

Frameworks & Libraries: React, React Native, Node.js, .NET, Flask, Django, TensorFlow, Scikit-Learn, Pandas, Numpy

Cloud & DevOps: AWS (Lambda, EC2, Glue), Google Cloud Platform (GCP), Azure, Docker, Kubernetes, Git, GitHub Actions

Tools & Concepts: RESTful APIs, Agile, Test-Driven Development, LLMs, Vector Databases (Pinecone), Spark, IDOR, SAST

EXPERIENCE

Software Engineer Intern

May 2026 – August 2026

Pinterest

New York, NY

- Built VulnScout, an AI-assisted application security agent that mines Pinterest's accepted Bugcrowd history to proactively surface variants of prior web/API authorization bugs (IDOR, broken access control) before external researchers report them.
- Engineered a LangGraph DeepAgent grounding LLM candidate generation in a vector DB of historical reports (multi-query RAG), Pinterest source code, and JIRA/GitHub fix-PR context via Semgrep-style searches over handlers and decorators.
- Designed a browser-based validation harness that reproduces candidates across 10+ sandboxed test-account sessions and routes validated/inconclusive/filtered findings to AppSec reviewers via Slack alerts; shipped on an internal tooling platform.
- Technologies:** TypeScript, React, Docker, MySQL, LangChain, RAG, LLMs, Semgrep, REST APIs, Browser Agents

Software Development Engineer Intern

February 2026 – April 2026

Amazon

Seattle, WA

- Designed and deployed multilingual personalization features for Amazon Music's voice recommendation system ($\approx 23\text{M}+$ daily requests), integrating user listening behavior into an ML ranking pipeline and achieving 96%+ feature coverage.
- Owned end-to-end system design and development of a language-aware candidate filtering system in Java, reducing irrelevant cross-language recommendations and improving music recommendation quality across 17M+ daily voice requests.
- Engineered 5 large-scale PySpark data pipelines (AWS Glue) to analyze 100M+ recommendation events, uncovering feature coverage gaps and critical quality issues that directly informed ranking model inputs and system design decisions.
- Technologies:** Java, PySpark, AWS (Glue), Reinforcement Learning (RL), A/B Testing, Distributed Systems

Software Engineer Intern

May 2025 – August 2025

Relativity

Chicago, IL

- Owned the end-to-end development of internal API extensions and automated GitHub Actions CI/CD workflows, reducing direct client data access during incident resolution by 20%+ and improving resolution speed.
- Designed and engineered fault-tolerant .NET (C#) migration jobs to transition over 1TB of data from legacy SQL systems to a distributed, Azure-hosted NoSQL architecture, enhancing horizontal scalability and reducing storage costs.
- Technologies:** C#, .NET, Azure (AKS), Kubernetes, Docker, MySQL, NoSQL, REST APIs, GitHub Actions

PROJECTS

PAUL LLM Assistant – Illinois Business Consulting (IBC) | Technical Lead

- Designed, developed, and containerized an internal LLM assistant using Python and Docker to enhance workflow efficiency for 250+ active consultants by intelligently answering complex organizational queries.
- Engineered a prompt classification system to route queries to either the OpenAI API or a Pinecone vector database, enabling semantic search across past project data to generate context-aware, case-relevant responses.
- Technologies:** Python, OpenAI API, Pinecone, Scikit-Learn, FastAPI, Docker, Large Language Models (LLMs)

Market Data Feeds Compression (PCAP Compression) – FinTech Lab @ UIUC | Software Engineer

- Designed and implemented a high-performance C++17 market data system for IEX TOPS/DEEP and NASDAQ ITCH PCAP feeds, using Zstandard (Zstd) with dictionary training to reduce data size by 80-85% ($\approx 15\%$ gain over generic Zstd).
- Engineered a streaming, memory-efficient architecture ($\leq 256\text{MB}$) capable of processing multi-GB captures at 40K+ packets/sec, suitable for continuous market-hours ingestion while maintaining fast decompression and system reliability.
- Technologies:** C++17, libpcap, Zstandard (Zstd), Make, Linux, Data Infrastructure, High Frequency Trading (HFT)

LEADERSHIP

Technology Director

May 2025 – December 2025

National Organization for Business and Engineering (NOBE): Illinois Chapter

Urbana-Champaign Area, IL

- Led a team of 40+ student software engineers across 4+ technical projects for a diverse portfolio of clients.